

SonicCraft

An Interactive Python Desktop Suite for Signal Analysis

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Part 1: System Architecture & Time-Domain

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Project Overview

- **Core Philosophy:** SonicCraft bridges foundational Signals & Systems theory with a real-time, responsive desktop interface.
- **Decoupled Architecture:** The PyQt6 GUI presentation layer is strictly separated from the non-blocking NumPy/SciPy DSP execution core.
- **Presentation Flow:** We will cover Time-Domain operations first, followed by advanced Frequency-Domain transformations and spectral editing.

Discrete Signal Manipulation

- Audio is loaded as discrete-time arrays via SoundFile.

Digital LTI Filtering

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- Amplitude scaling and envelope fading via element-wise array multiplication.

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Time-Domain Operations & LTI Systems

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- 9-Band Parametric EQ and standard IIR/FIR filters (`SciPy`).

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- Represented by the linear difference equation:

$$\sum_{k=0}^N a_k y[n - k] = \sum_{m=0}^M b_m x[n - m]$$

Part 2: Frequency-Domain & Spectral Editing

Presenter: Mahbubul Islam Mahi

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Fast Fourier Transform (FFT)

Converts continuous-time theory into discrete computational bins to identify dominant frequencies in real-time.

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Short-Time Fourier Transform (STFT)

- Chops audio into windowed, overlapping blocks to create a 2D matrix (Time vs. Frequency vs. Magnitude).
- Visualized in the UI as an interactive Spectrogram heatmap.

- **Spectral Masking:** Users can draw a bounding box over specific time/frequency coordinates on the Spectrogram to mathematically zero-out unwanted frequencies.

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- **Image-to-Audio Reconstruction:** Uses PIL to read grayscale images, map pixels to frequency bins, and apply the Griffin-Lim algorithm to synthesize playable audio arrays.
- **Inverse STFT:** Both processes utilize the Inverse STFT to rebuild modified magnitude spectra back into a 1D time-domain waveform.

[Insert Live UI Screenshot Here]

Demonstrating the resizable QSplitter layout with the Waveform on top and the Spectrogram analyzer below.

Thank You!

Questions & Answers